

Please solve the following exercises at home prior to the tutorial

For the next two exercises we'll be looking at the following elliptic curve over the finite field with 11 elements: $y^2 = x^3 + 2x + 10$

Exercise 1

- a) Does the above equation satisfy the conditions for an elliptic curve?
- b) Determine all points (x, y) on this curve
(Hint: There's a total of 7 points, not counting the neutral element).

The following exercises will be done during the tutorial:

Exercise 2

Perform a Diffie-Hellman key exchange using the above curve.

- a) Which of the following points can in principle be used as a public element g ? Which one is secure, i.e., which one is a primitive element?
 $(1, 2), (7, 2), (9, 3)$
- b) The public generator is $g = (4, 7)$. Alice chooses the secret number $x_A = 3$, Bob chooses $x_B = 6$. Determine the shared key by performing the calculations for both sides (Alice und Bob).